

COOPER DAVIES, PHD

📞 1-403-861-2722 @ coopertsd@gmail.com 📍 Calgary, Alberta, CAN
in <https://www.linkedin.com/in/cooper-davies-phd-91bba9155/> 🐙 <https://github.com/DaviesCooper> 🌐 <https://wildroseplains.ca>

PROFILE

Senior AI Engineer and technical lead with 8+ years delivering applied ML, conversational AI, and full-stack products. Ph.D., M.Sc., and B.Sc. Computer Science all from the University of Calgary. Experience spans rapid prototyping and production systems: leading teams, owning client delivery, and shipping under budget and time constraints.
Technical: Python, C#, JavaScript/TypeScript, React Native, Azure, GCP, Firebase, PostgreSQL, LLMs, AI/ML, AutoML, RAG, vector DBs, YOLO/OCR, Unity, VR/MR/XR.

EXPERIENCE

Senior AI Engineer

Remote Inc. • Permanent Full-time

📅 Aug 2024 – Present 📍 Remote - California, USA

- Architected and launched a full-stack H5 game platform (React Native + Expo + custom Game SDK), reducing game development time by upwards of 95% and enabling non-technical users to ship games with zero code
- Built a game management platform (GCP Cloud Run + Firebase) supporting 2,000+ active users and reducing operational overhead by three quarters through automation
- Integrated 6+ external systems (Firebase, AppsFlyer, Cloudinary, GitHub Actions, Algolia, Supabase), improving data flow reliability and reducing manual intervention by half
- Owned and maintained the company-wide Universal LLM Server, cutting LLM integration time (approximately 80% faster) and reducing API costs by 25–35% via centralized token tracking and optimization
- Developed and standardized an internal agentic AI framework using RAG + vector databases (Databricks, PostgreSQL), improving response relevance by 30%+ and enabling scalable knowledge retrieval across all company documents

Senior Software Lead

VizworX Inc. • Permanent Full-time

📅 Apr 2018 – Jan 2024 📍 Calgary, Alberta, Canada

- Led and scaled multiple engineering pods (3–6 engineers each) across applied AI and XR projects, consistently delivering on parallel project timelines and improving team throughput by 30%
- Owned end-to-end client delivery, defining product roadmaps with SMEs, removing daily engineering blockers, and enforcing release quality standards, resulting in on-time delivery across 10+ projects
- Co-architected and launched the Panoptica VR industrial design platform (Microsoft Azure), enabling immersive design workflows and reducing design iteration cost by multiple fractions
- Developed a recommendation engine for a niche oil & gas e-commerce platform, increasing specialized part adoption by 25% and improving user engagement
- Built a high-performance drill tenacity prediction system using a custom C4.5-style decision tree compiled to native x86, enabling real-time predictions on embedded hardware with minimal latency

Data Scientist

RetinaLogik • Contract Part-time

📅 Jan 2023 – Dec 2023 📍 Calgary, Alberta, Canada

- Reverse-engineered and optimized the Swedish Interactive Thresholding Algorithm (SITA), reducing glaucoma visual field test duration from 10 minutes to 1.5 minutes while maintaining clinical accuracy
- Integrated the optimized testing algorithm into an XR-based diagnostic platform on PICO devices, enabling real-time patient testing and improving usability in clinical environments

EDUCATION

Ph.D. in Computer Science: AutoML
University of Calgary

📅 Class of 2024

M.Sc. in Computer Science: AutoML
University of Calgary

📅 Class of 2020

B.Sc. in Computer Science
University of Calgary

📅 Class of 2017

WEBSITE

[Wildroseplains.ca](https://wildroseplains.ca)

My website contains a portfolio of my work. It's also an interactive linux terminal. If you're even slightly interested in Linux I suggest giving it a look.

Machine Learning Lead Developer

ReVisionz Inc. • Contract Part-time

📅 Aug 2021 – Dec 2021

📍 Calgary, Alberta, Canada

- Led a 10-person engineering team to design and deliver a distributed OCR pipeline processing 5M+ oil & gas engineering drawings, achieving 95% usable data extraction from low-quality inputs
- Architected and deployed a scalable Azure-based processing system (VMs + Blob Storage), increasing parallel throughput by 5–10x and enabling efficient large-scale document processing
- Implemented robust monadic error-handling across the pipeline, ensuring 100% document traceability and visibility regardless of processing failures
- Improved text extraction accuracy by replacing Tesseract with Azure OCR and integrating YOLO/Darknet for bounding-box segmentation, significantly increasing structured data quality and downstream usability